Random Forest Assignment

1. To predict insurance charges based on the Input

2. The Basic information about the dataset.

Total Number of rows: 1339 Total number of columns:5

Sl.No	N_estimators	Criterion	Max_features	R Value
1	50	Squared_error	1.0	0.8603
2	50	Friedman_mse	None	0.8601
3	50	Poisson	1.0	0.8548
4	50	Absolute_error	None	0.8523
3	100	Squared_error	1.0	0.8543
4	100	Friedman_mse	None	0.8536
5	100	Poisson	1.0	0.8564
6	100	Absolute_error	None	0.8545
7	150	Squared_error	1.0	0.8587
8	150	Friedman_mse	None	0.8584
9	150	Poisson	1.0	0.8542
10	150	Squared_error	1.0	0.8619
11	150	Absolute_error	None	0.8564
12	150	Friedman_mse	1.0	0.8567
13	100	Squared error	1.0	0.8617
14	100	Friedman_mse	None	0.8578
15	100	Poisson	None	0.8560
16	100	Absolute_error	None	0.8588

The Best model comes with no of estimators as 150 and criterion in Squared_error and max_Features is 1.0

The R value is 86.19